

MCMC Samplers for the Nuclear-Norm Distribution Posterior

Notes on Segert & Wycoff (2025), arXiv:2510.05447

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Source paper

Segert, J. & Wycoff, N. (2025). *A Probabilistic Basis for Low-Rank Matrix Learning*. arXiv:2510.05447v1.

Model

Observation model (heteroscedastic; entries missing when $s_{ij} = \infty$):

$$y_{ij} \mid x_{ij}, s_{ij} \sim \mathcal{N}(x_{ij}, s_{ij}^2), \quad (i, j) \in \Omega.$$

Nuclear-norm prior with scalar $\lambda > 0$ (Proposition 3.3 of source):

$$p(X \mid \lambda) \propto \lambda^{mn} \exp(-\lambda \|X\|_*).$$

Half-Cauchy hyperprior on λ (Section 4.3 of source; Polson & Scott 2012): implemented via the inverse-Gamma hierarchy of Makalic & Schmidt (2015)—see Section below.

SVD reparameterization

Write the thin SVD $X = U \text{diag}(\sigma) V^T$ where $U \in \text{St}(m, n)$ (semi-orthogonal), $V \in O(n)$ (orthogonal), $\sigma \in \mathbb{R}_+^n$. The nuclear norm is $\|X\|_* = \sum_k \sigma_k$. Reparameterizing to (U, σ, V) gives tractable conditional posteriors (Section 4.2 of source; homoscedastic $s_{ij}^2 = \gamma^2$).

Sampler 1: Proximal MALA (Sec. 4.1, Appendix J.4)

Algorithm. Metropolis-Adjusted Langevin Algorithm using the proximal operator of $-\log p$ as the proposal mean. For the heteroscedastic setting, the proximal operator of $\lambda \|\cdot\|_*$ does not simplify to a closed-form expression (unlike the homoscedastic case in Appendix J.3 where the proximal operator of the negative log-posterior has a closed form via SVT). We therefore use a *proximal gradient* step instead (as in the matrix completion case, Appendix J.4):

$$\bar{X} = \text{SVT}_{\delta\lambda/2} \left(X^t + \frac{\delta}{2} \underbrace{M \odot (Y - X^t) / S^2}_{\nabla_X \log p_{\text{lik}}(X^t)} \right),$$

where $M_{ij} = \mathbf{1}[(i, j) \in \Omega]$, $\text{SVT}_\tau(Z) = U \text{diag}(\max(\sigma_k - \tau, 0)) V^T$ is singular-value soft-thresholding, and $\delta > 0$ is the proposal scale.

Proposal and accept/reject.

$$X^* \sim \mathcal{N}(\bar{X}, \delta I),$$

accepted with probability $\min(1, \alpha)$ where

$$\log \alpha = \log p(X^*) - \log p(X^t) + \log q(X^t | X^*) - \log q(X^* | X^t),$$

$$\log p(Z) = -\frac{1}{2} \sum_{(i,j) \in \Omega} \frac{(y_{ij} - z_{ij})^2}{s_{ij}^2} - \lambda \|Z\|_*,$$

$$\log q(Z' | Z) = -\frac{\|Z' - \bar{Z}\|_F^2}{2\delta},$$

with $\bar{Z}$ the proximal-gradient mean centred at Z .

Step size adaptation. Robbins–Monro targeting acceptance rate 0.574 (optimal for proximal MCMC; Crucinio et al. 2023):

$$\log \delta_{t+1} = \log \delta_t + c_t(\alpha_t - 0.574), \quad c_t = c/\sqrt{t}.$$

Heteroscedastic adaptation. In the homoscedastic case (Appendix J.3), the proximal operator of the full negative log-posterior has the closed form $\text{SVT}_{\delta\lambda(\delta+2\gamma^2)^{-1}}((\delta Y + 2\gamma^2 X^t)(\delta + 2\gamma^2)^{-1})$. For heteroscedastic s_{ij}^2 , no such closed form exists; the proximal gradient approximation above is used instead, exactly as in the matrix completion variant of Appendix J.4. The MH correction renders the chain exact regardless of this approximation.

Sampler 2: SVD Gibbs (Sec. 4.2)

Algorithm. Iterate over three conditional samplers in the SVD parameterization $X = U \text{diag}(\sigma) V^T$. Let $s = \text{diag}(U^T Y V)$.

Step A: Sample $\sigma | U, V, \lambda, Y$. The joint conditional density of the singular values is

$$p(\sigma | \cdot) \propto \exp\left(-\lambda \sum_k \sigma_k - \frac{\|\sigma - s\|^2}{2\gamma^2}\right) \prod_{k \leq n} \sigma_k^{m-n} \prod_{i < j \leq n} |\sigma_i^2 - \sigma_j^2|,$$

where the Vandermonde factor $\prod_{i < j} |\sigma_i^2 - \sigma_j^2|$ arises from the Jacobian of the SVD map. This density is not product-separable and has posterior nonsmoothness from the nuclear norm; the source samples it via a Metropolis-adjusted Langevin method operating in $\mathbb{R}^n$ (vs. the full $m \times n$ problem for direct MALA on X).

Step B: Sample $U | \sigma, V, \lambda, Y$.

$$p(U | \cdot) \propto \exp\left(-\frac{1}{2\gamma^2} \text{tr}[(YV \text{diag}(\sigma))^T U]\right),$$

which is a *matrix von Mises–Fisher* (vMF) distribution on $\text{St}(m, n)$ (Jupp & Mardia 1979). Sampled via the reflective slice sampler of Hoff (2009).

Step C: Sample $V \mid \sigma, U, \lambda, Y$.

$$p(V \mid \cdot) \propto \exp\left(-\frac{1}{2\gamma^2} \text{tr}[(\text{diag}(\sigma)Y^T U^T)^T V]\right),$$

again a matrix vMF on $O(n)$; sampled via Hoff (2009).

The variational bound is not a scale mixture. Proposition 3.2 of the source gives a variational (minimax) identity:

$$\exp(-\lambda \|X\|_*) = \min_{Q \succ 0} \exp\left(-\frac{\lambda}{2} [\text{tr}(Q^{-1}\Psi) + \text{tr}(Q)]\right), \quad \Psi = X^T X,$$

minimized at $Q_* = \Psi^{1/2}$ by the AM–GM inequality $\text{tr}(Q^{-1}\Psi) + \text{tr}(Q) \geq 2 \text{tr}(\Psi^{1/2}) = 2\|X\|_*$. This is a *minimization*, not a marginalization: the *integral* $\int_{Q \succ 0} \exp\left(-\frac{\lambda}{2} [\text{tr}(Q^{-1}\Psi) + \text{tr}(Q)]\right) dQ$ does *not* equal $\exp(-\lambda \|X\|_*)$ in general. A Gibbs chain that treats Q as an auxiliary variable and samples from the “pseudo-joint” $p(X, Q) \propto \exp\left(-\frac{\lambda}{2} [\text{tr}(Q^{-1}\Psi) + \text{tr}(Q)]\right) \cdot p(Y|X)$ therefore targets the *wrong* posterior.

Correct Gaussian scale mixture. The proper augmentation uses the modified Bessel identity $K_{1/2}(z) = \sqrt{\pi/(2z)} e^{-z}$:

$$\int_0^\infty d^{-1/2} \exp\left(-\frac{\psi}{2d} - \frac{\lambda^2 d}{2}\right) dd = \frac{\sqrt{\pi}}{\lambda} e^{-\lambda \sqrt{\psi}} = \frac{\sqrt{\pi}}{\lambda} e^{-\lambda \sigma}, \quad \sigma = \sqrt{\psi},$$

so that

$$\exp(-\lambda \|X\|_*) = \exp(-\lambda \sum_k \sigma_k) \propto \int_{\mathbb{R}_{>0}^n} \prod_k d_k^{-1/2} \exp\left(-\frac{\psi_k}{2d_k} - \frac{\lambda^2 d_k}{2}\right) \prod_k dd_k,$$

where ψ_k are the eigenvalues of $\Psi = X^T X$ (i.e. $\psi_k = \sigma_k^2$). Crucially, λ appears *quadratically* in the exponential (not linearly as in the variational bound). The auxiliary variable Q must share eigenvectors with Ψ – write $Q = V \text{diag}(d) V^T$ where $\Psi = V \text{diag}(\psi) V^T$ – for the integral to factorise. The correct conditional is

$$p(d_k \mid X, \lambda) \propto d_k^{-1/2} \exp\left(-\frac{\psi_k}{2d_k} - \frac{\lambda^2 d_k}{2}\right) = \text{GIG}\left(\frac{1}{2}, \psi_k, \lambda^2\right).$$

This differs from the variational-bound “conditional” in two ways: the shape parameter is $\frac{1}{2}$ (giving the $d_k^{-1/2}$ factor) and the rate on d_k is $\lambda^2/2$, not $\lambda/2$.

Heteroscedastic Gibbs (corrected). For heteroscedastic s_{ij}^2 , the vMF structure of the conditionals for U and V is lost. Using the correct GSM with auxiliary d in the eigenbasis of $X^T X$, the three-step Gibbs is:

Step A: Sample $X \mid d, \lambda, Y, S$. Writing $p(x_i \mid Q, \lambda) \propto \exp(-\frac{1}{2} x_i^T Q^{-1} x_i)$ with precision λQ^{-1} , each row is a Gaussian:

$$A_i = \text{diag}(s_{i,\cdot}^{-2} \odot m_{i,\cdot}) + \lambda Q^{-1}, \quad r_i = (s_{i,\cdot}^{-2} \odot m_{i,\cdot}) \odot y_i^{\text{obs}}, \quad (1)$$

$$x_i \sim \mathcal{N}(A_i^{-1} r_i, A_i^{-1}), \quad (2)$$

where $Q^{-1} = V \text{diag}(1/d) V^T$. Solved via Cholesky of the $n \times n$ precision A_i ; rows parallelised via vmap.

Step B: Sample $d \mid X, \lambda$ (exact GIG). After sampling X_{new} , eigendecompose $\Psi_{\text{new}} = V_{\text{new}} \text{diag}(\psi_{\text{new}}) V_{\text{new}}^T$. Each d_k is independently drawn from

$$d_k \mid \psi_k, \lambda \sim \text{GIG}\left(\frac{1}{2}, \psi_k, \lambda^2\right),$$

with density $\propto d_k^{-1/2} \exp(-\psi_k/(2d_k) - \lambda^2 d_k/2)$. GIG samples can be drawn exactly (no Metropolis step needed) using the algorithm of Devroye (2014) or Hörmann & Leydold (2014); see Appendix. The mode is $d_k^* = (-\frac{1}{2} + \sqrt{\frac{1}{4} + \lambda^2 \psi_k})/\lambda^2$ (close to σ_k/λ for large $\lambda\sigma_k$).

Step C: Sample $\lambda \mid d_{\text{new}}$ (half-Cauchy hierarchy). From the correct GSM joint, $p(\lambda \mid d) \propto \lambda^{(m+1)n} \exp(-\frac{\lambda^2}{2} \sum_k d_k) \cdot p(\lambda)$, where the $\lambda^{(m+1)n}$ factor accounts for the λ^{mn} normalising constant of the NND prior and the λ^n from the GIG normalising constants. Adopting a half-Cauchy prior on λ via the half-Normal scale mixture (Polson & Scott 2012): $\lambda|\zeta \sim \text{half-Normal}(0, \zeta)$, $\zeta \sim \text{IG}(\frac{1}{2}, \frac{1}{2})$:

$$\zeta' \sim \text{IG}\left(\frac{1}{2}, \frac{1+\lambda^2}{2}\right), \quad (3)$$

$$\lambda' \sim \text{TruncatedNormal}^+\left(0, [\sum_k d'_k + 1/\zeta']^{-1}\right), \quad (4)$$

where $\text{TruncatedNormal}^+(0, v)$ denotes a half-normal with variance v , sampled as $|\mathcal{N}(0, 1)|/\sqrt{\sum d'_k + 1/\zeta'}$. The $(m+1)n$ exponent modifies (4) to a truncated generalised normal; if $(m+1)n$ is large this can be approximated or handled via MH.

Comparison with current MATLAB implementation. The current code uses the variational pseudo-joint rather than the correct scale mixture, giving two errors in Step B: (i) the GIG shape is 1 instead of $\frac{1}{2}$ (missing the $d_k^{-1/2}$ factor), and (ii) the rate on d_k is $\lambda/2$ instead of $\lambda^2/2$. Step C also uses the $\exp(-\lambda \text{tr}(Q))$ form (linear in λ) rather than the quadratic-in- λ form from the correct scale mixture. The sampler is therefore approximate; the stationary distribution differs from the true posterior.

Sparse-data behaviour. At high missing fractions ($\gtrsim 60\%$), $X_{\text{new}} \rightarrow 0$ under the posterior, so $\psi_k \rightarrow 0$, $\sum d_k \rightarrow 0$ and $\lambda' \rightarrow \infty$. This occurs with both the variational and the correct GSM formulations: the NND prior places all mass at $X = 0$ for large λ , and sparse data cannot rule out this degenerate configuration.

λ sampling: Half-Cauchy prior (Sec. 4.3)

A Gamma prior on λ is conjugate but may not allow λ to range over multiple orders of magnitude. The source uses a *half-Cauchy* prior (Polson & Scott 2012), implemented via the inverse-Gamma hierarchy of Makalic & Schmidt (2015): if $\alpha|\beta \sim \text{IG}(1/2, 1/\beta)$ and $\beta \sim \text{IG}(1/2, 1)$ then α is marginally Half-Cauchy. For conjugacy with the NND likelihood $p(X|\lambda) \propto \lambda^{mn} \exp(-\lambda \|X\|_*)$, the source samples the *inverse* $\alpha = 1/\lambda$ and

introduces an auxiliary β . In the SVD Gibbs (homoscedastic), the conjugate updates are:

$$\beta|\alpha \sim \text{IG}\left(1, 1 + \frac{1}{\alpha}\right), \quad (5)$$

$$\alpha|\beta, X, \gamma^2 \sim \text{IG}\left(mn + \frac{1}{2}, \frac{\|X\|_*}{\gamma^2} + \frac{1}{\beta}\right), \quad (6)$$

$$\lambda \leftarrow \frac{1}{\alpha}. \quad (7)$$

Note on parameterisations. The IG hierarchy above is conjugate to the *linear* NND likelihood $\exp(-\lambda\|X\|_*/\gamma^2)$ (source uses homoscedastic γ^2). In the correct GSM augmentation (Section), λ appears *quadratically* via $\exp(-\lambda^2 \sum_k d_k/2)$, so the conditional $p(\lambda|d)$ has a different form (see Step C in Section) and the IG hierarchy no longer applies. The current MATLAB implementation applies the IG hierarchy with $\text{tr}(Q) = \sum_k d_k$ in place of $\|X\|_*/\gamma^2$; this is internally consistent with the (approximate) variational pseudo-joint but not with the correct GSM.